# MAHMOOD TAUHIDUL

Software Engineer

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## ABOUT ME

Software engineer experienced in end-to-end delivery of desktop applications, browser extensions, learning platforms and healthcare systems. Work includes software architecture, backend and frontend development, authentication, databases, testing, CI/CD and cloud deployments. Technical background spans Go, React, TypeScript, C++, Swift, Python, GCP and AWS.

## WORK EXPERIENCE

**Software Engineer | Guided Doing** 03/2026 – Present

Langen (Hessen), Germany · Remote

- Owned projects across requirements, architecture, implementation, testing, deployment and maintenance; designed database models, authentication flows and notification streams.
- Built workflow-capture browser extension and Windows/macOS companion desktop applications with C++ and Swift integrations.
- Developed multi-academy learning management system using Go backend and React frontend, with element-based lessons, AI integrations and academy-specific subdomains.
- Implemented Firebase Cloud Functions and backend integrations; managed GCP, CI/CD, automated multi-environment deployment and release workflows.

**Full Stack Developer | BinduLogic Limited** 09/2025 – 03/2026

Dhaka, Bangladesh · 17 September 2025 – 24 March 2026

- Built hospital portals and healthcare platforms with Next.js, React, TypeScript, Payload CMS and PostgreSQL.
- Created reusable content blocks, English/Bengali localization, live preview and cross-collection search for clinical and administrative content.
- Integrated AWS S3/CloudFront; supported database migrations, role-based access, accessibility and responsive interfaces.

**Software Engineer Intern | BinduLogic Limited** 03/2025 – 09/2025

Dhaka, Bangladesh · 16 March – 16 September 2025

- Contributed to frontend and backend development using TypeScript, JavaScript, React, Next.js and Node.js; participated in code reviews and technical discussions.

**Software Intern | Al Khamis International** 01/2023 – 10/2023

Saudi Arabia · Remote

- Built passport-tracking, employee and expense management tools using Node.js, Express and MySQL.
- Automated government-portal data extraction with Python, Selenium and BeautifulSoup.

## EDUCATION AND TRAINING

**B.Sc. in Computer Science & Engineering | Independent University, Bangladesh** 2020 – 2025

Dhaka, Bangladesh · Minor: Marketing · CGPA: 3.07/4.00

- Selected coursework: Object-Oriented Programming (A), Algorithms (A), Systems Analysis and Design (A), Web Applications and Internet (A), Image Processing (B−), Probability & Statistics (B+), Linear Algebra (B).

## PROJECTS

**Fabric Classification & GSM Prediction | Image Processing** 2025

Academic course project

- Developed CNN-based textile image classification and GSM regression using VGG16/ResNet50, Python, TensorFlow/PyTorch and OpenCV.
- Prepared 1,000+ labeled fabric samples, performed image preprocessing and augmentation, and built a Flask inference API. Reported project results: 99% classification accuracy and $R^2$ of 0.95.

**SizeLens — Automated Measurement | Microprocessor** 2023

Academic course project

- Built ESP32-CAM prototype with YOLOv3 and OpenCV for camera calibration, perspective correction, contour analysis and physical measurement; reported approximately 2 mm accuracy.
- Created a Flask interface for calibration, measurements and captured-image review.

**UniPlan — University Degree Planner** 2024

Full-stack academic planning application

- Built scheduling, prerequisites, GPA calculation, forums and retrieval-augmented course recommendations.
- Implemented JWT/RBAC, WebSocket notifications, drag-and-drop scheduling and PostgreSQL optimization using Next.js/React and Node.js.

## RESEARCH MANUSCRIPTS — SUBMITTED FOR PUBLICATION

The following manuscripts have been submitted; acceptance and publication are not claimed.

- IoT-Based Body Size Measurement System
- Deep Learning for Fabric Classification
- Textile Fabric Dataset for Machine Learning
- GSM Prediction Using Convolutional Neural Networks

## LANGUAGE SKILLS

Mother tongue: Bengali | Other language: English — C1 (self-reported; IELTS overall score 7 previously reported)

## DIGITAL AND TECHNICAL SKILLS

Programming: Go, Python, C++, Swift, TypeScript, JavaScript, Rust, Java, SQL

Applications & backend: React, Next.js, Node.js, Express, REST/GraphQL APIs, Payload CMS, Firebase Cloud Functions

Data, cloud & delivery: PostgreSQL, MySQL, GCP, AWS, Docker, GitHub Actions, CI/CD, automated deployments

Machine learning: PyTorch, TensorFlow, Keras, OpenCV, YOLO, CNNs, transfer learning, RAG

## OPEN SOURCE AND ADDITIONAL TRAINING

Contributions: GraphBit (Rust asynchronous testing/CI), Infinitibit Document Parser (Python), OpenNotebook and OpenCut (testing/documentation).

Training: CS50: Introduction to Computer Science (Harvard); Networking Basics (Cisco Networking Academy).